

Panitumumab in combination with infusional oxaliplatin and oral capecitabine for conversion therapy in patients with colon cancer and advanced liver metastases. The MetaPan study.

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Preoperative chemotherapy improves the outcome in patients with colorectal cancer with liver metastases. In the current study, the authors evaluated the activity of a conversion treatment with the combination of capecitabine plus oxaliplatin (XELOX) used in association with panitumumab in patients with unresectable, liver-only, metastatic colon cancer. Chemotherapy-naïve patients with unresectable liver metastases from colon cancer with no other metastatic disease sites were enrolled. All patients received upfront therapy with XELOX plus panitumumab (P-XELOX) and were reevaluated for resectability every 4 cycles. The primary endpoint was the objective response rate (ORR). Secondary endpoints were overall survival (OS), progression-free survival, the percentage of patients whose disease became radically resectable, and the safety of the P-XELOX combination. A total of 49 patients were recruited, 35 of whom had wild-type KRAS (wtKRAS) and 14 of whom (who were enrolled before study amendment) had unknown (9 patients) or mutated (5 patients) KRAS mutational status. Forty-six patients were evaluable for response. After conversion P-XELOX therapy, the ORR in the general population was 54%, with 2 complete responses, 23 partial responses, and 14 cases of stable disease. In patients with wtKRAS, the ORR of the patients reached 65% (2 CRs and 19 PRs), which allowed 15 patients with initial unresectable liver metastasis to be reclassified as having resectable disease. Survival analysis demonstrated a median progression-free survival of 8.5 months and a median OS of 21.9 months. Patients who underwent surgery were found to have a significantly better OS when compared with those who did not undergo surgery ($P < .001$). Overall, toxicities were found to be predictable and manageable, with the most common being cutaneous, gastrointestinal, and neurologic toxicities. Conversion P-XELOX therapy yields high response and resectability rates for patients with metastatic colon cancer with extensive liver involvement.

Liver-first approach to the treatment of patients with synchronous colorectal liver metastases: a systematic review and meta-analysis.

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The optimal approach to the treatment of colorectal carcinoma and synchronous liver metastases remains controversial. The objective of this review was to analyze the outcomes of adopting the liver-first approach for the treatment of patients with colorectal cancer with synchronous hepatic metastases who initially underwent systemic chemotherapy and/or resection of the metastatic lesions and primary colorectal carcinoma. This review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. The MEDLINE, EMBASE, LILACS, and Cochrane Central Register of Controlled Trials databases were searched for the identification and retrieval of eligible studies. Studies that included details of using the liver-first approach for the treatment of synchronous liver metastases of colorectal cancer and its outcomes, including the patients' survival data, were included. Proportional meta-analysis was performed using the random-effects restricted maximum likelihood method to summarize the three- and five-year overall survival and recurrence rates of the patients. Eight hundred and fifty-five articles describing the results of studies on the liver-first approach were identified. Three independent reviewers screened the titles and abstracts of the articles and excluded 750 articles. Thereafter, 29 retrospective and comparative studies that met the inclusion

criteria were included. No randomized controlled trials were identified in the database search. Neoadjuvant treatment with systemic chemotherapy for hepatic metastasis can prepare a patient for resection of liver metastases, offering the opportunity for potentially curative treatment of synchronous hepatic metastases initially considered unresectable. The decision regarding the resection of primary colorectal carcinoma and liver metastases should be based on individualized patient response. Prospero database registration ID: CRD42022337047 (www.crd.york.ac.uk/prospero).

Conversion to resectability using hepatic artery infusion plus systemic chemotherapy for the treatment of unresectable liver metastases from colorectal carcinoma.

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PURPOSE To determine the conversion to resectability in patients with unresectable liver metastases from colorectal cancer treated with hepatic arterial infusion (HAI) plus systemic oxaliplatin and irinotecan (CPT-11). **PATIENTS AND METHODS** Forty-nine patients with unresectable liver metastases (53% previously treated with chemotherapy) were enrolled onto a phase I protocol with HAI floxuridine and dexamethasone plus systemic chemotherapy with oxaliplatin and irinotecan. **Results** Ninety-two percent of the 49 patients had complete (8%) or partial (84%) response, and 23 (47%) of the 49 patients were able to undergo resection in a group of patients with extensive disease (73% with > five liver lesions, 98% with bilobar disease, 86% with > or = six segments involved). For chemotherapy-naïve and previously treated patients, the median survival from the start of HAI therapy was 50.8 and 35 months, respectively. The only baseline variable significantly associated with a higher resection rate was female sex. Variables reflecting extensive anatomic disease, such as number of lesions or number of vessels involved, were not significantly associated with the probability of resection. **CONCLUSION** The combination of regional HAI floxuridine/dexamethasone and systemic oxaliplatin and irinotecan is an effective regimen for the treatment of patients with unresectable liver metastases from colorectal cancer, demonstrating a 47% conversion to resection (57% in chemotherapy-naïve patients). Future randomized trials should compare HAI plus systemic chemotherapy with systemic therapy alone to assess the additional value of HAI therapy in converting patients with hepatic metastases to resectability.

First-line systemic treatment strategies in patients with initially unresectable colorectal cancer liver metastases (CAIRO5): an open-label, multicentre, randomised, controlled, phase 3 study from the Dutch Colorectal Cancer Group.

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Patients with initially unresectable colorectal cancer liver metastases might qualify for local treatment with curative intent after reducing the tumour size by induction systemic treatment. We aimed to compare the currently most active induction regimens. In this open-label, multicentre, randomised, phase 3 study (CAIRO5), patients aged 18 years or older with histologically confirmed colorectal cancer, known RAS/BRAFV600E mutation status, WHO performance status of 0-1, and initially unresectable colorectal cancer liver metastases were enrolled at 46 Dutch and one Belgian secondary and tertiary centres. Resectability or unresectability of colorectal cancer liver metastases was assessed centrally by an expert panel of liver surgeons and radiologists, at baseline and every 2 months thereafter by predefined criteria. Randomisation was done centrally with the minimisation technique via a masked web-based allocation

procedure. Patients with right-sided primary tumour site or RAS or BRAFV600E mutated tumours were randomly assigned (1:1) to receive FOLFOX or FOLFIRI plus bevacizumab (group A) or FOLFOXIRI plus bevacizumab (group B). Patients with left-sided and RAS and BRAFV600E wild-type tumours were randomly assigned (1:1) to receive FOLFOX or FOLFIRI plus bevacizumab (group C) or FOLFOX or FOLFIRI plus panitumumab (group D), every 14 days for up to 12 cycles. Patients were stratified by resectability of colorectal cancer liver metastases, serum lactate dehydrogenase concentration, choice of irinotecan versus oxaliplatin, and BRAFV600E mutation status (for groups A and B). Bevacizumab was administered intravenously at 5 mg/kg. Panitumumab was administered intravenously at 6 mg/kg. FOLFIRI consisted of intravenous infusion of irinotecan at 180 mg/m² with folinic acid at 400 mg/m², followed by bolus fluorouracil at 400 mg/m² intravenously, followed by continuous infusion of fluorouracil at 2400 mg/m². FOLFOX consisted of oxaliplatin at 85 mg/m² intravenously together with the same schedule of folinic acid and fluorouracil as in FOLFIRI. FOLFOXIRI consisted of irinotecan at 165 mg/m² intravenously, followed by intravenous infusion of oxaliplatin at 85 mg/m² with folinic acid at 400 mg/m², followed by continuous infusion of fluorouracil at 3200 mg/m². Patients and investigators were not masked to treatment allocation. The primary outcome was progression-free survival, analysed on a modified intention-to-treat basis, excluding patients who withdrew consent before starting study treatment or violated major entry criteria (no metastatic colorectal cancer, or previous liver surgery for colorectal cancer liver metastases). The study is registered with ClinicalTrials.gov, NCT02162563, and accrual is complete. Between Nov 13, 2014, and Jan 31, 2022, 530 patients (327 [62%] male and 203 [38%] female; median age 62 years [IQR 54-69]) were randomly assigned: 148 (28%) patients to group A, 146 (28%) patients to group B, 118 (22%) patients to group C, and 118 (22%) patients to group D. Groups C and D were prematurely closed for futility. 521 patients were included in the modified intention-to-treat population (147 in group A, 144 in group B, 114 in group C, and 116 in group D). The median follow-up at the time of this analysis was 51.1 months (95% CI 47.7-53.1) in groups A and B and 49.9 months (44.5-52.5) in groups C and D. Median progression-free survival was 9.0 months (95% CI 7.7-10.5) in group A versus 10.6 months (9.9-12.1) in group B (stratified hazard ratio [HR] 0.76 [95% CI 0.60-0.98]; $p=0.032$), and 10.8 months (9.9-12.6) in group C versus 10.4 months (9.8-13.0) in group D (stratified HR 1.11 [95% CI 0.84-1.48]; $p=0.46$). The most frequent grade 3-4 events in groups A and B were neutropenia (19 [13%] patients in group A vs 57 [40%] in group B; $p<0.0001$), hypertension (21 [14%] vs 20 [14%]; $p=1.00$), and diarrhoea (five [3%] vs 28 [19%]; $p<0.0001$), and in groups C and D were neutropenia (29 [25%] vs 24 [21%]; $p=0.44$), skin toxicity (one [1%] vs 29 [25%]; $p<0.0001$), hypertension (20 [18%] vs eight [7%]; $p=0.016$), and diarrhoea (five [4%] vs 18 [16%]; $p=0.0072$). Serious adverse events occurred in 46 (31%) patients in group A, 75 (52%) patients in group B, 41 (36%) patients in group C, and 49 (42%) patients in group D. Seven treatment-related deaths were reported in group B (two due to multiorgan failure, and one each due to sepsis, pneumonia, portal vein thrombosis, septic shock and liver failure, and sudden death), one in group C (multiorgan failure), and three in group D (cardiac arrest, pulmonary embolism, and abdominal sepsis). In patients with initially unresectable colorectal cancer liver metastases, FOLFOXIRI-bevacizumab was the preferred treatment in patients with a right-sided or RAS or BRAFV600E mutated primary tumour. In patients with a left-sided and RAS and BRAFV600E wild-type tumour, the addition of panitumumab to FOLFOX or FOLFIRI showed no clinical benefit over bevacizumab, but was associated with more toxicity. Roche and Amgen.

Triplet Chemotherapy (FOLFOXIRI) Plus Bevacizumab Versus Doublet Chemotherapy (FOLFOX/FOLFIRI) Plus Bevacizumab in Conversion Therapy for Metastatic Colorectal Cancer: a Meta-Analysis.

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Conversion therapy can convert unresectable metastatic colorectal cancer (mCRC) into resectable. However, the optimal conversion regimen was not yet defined. This meta-analysis aimed to compare the efficacy and safety of the triplet chemotherapy (FOLFOXIRI) plus bevacizumab (Bev) with doublet chemotherapy (FOLFOX/FOLFIRI) plus Bev in conversion therapy. Randomized controlled trials (RCTs) from databases, including Pubmed, EMBASE, Cochrane clinical trials, clinicaltrial.gov and some conferences, were searched from the inception to November 2017. The R0 resection, objective response rate (ORR), progression-free survival (PFS), overall survival (OS) and the incidence of adverse events were pooled with the use of hazard ratio (HR) or risk ratio (RR). Four RCTs with 1013 patients were included. FOLFOXIRI plus Bev regimen significantly improved the overall R0 resection rate (RR 1.41, 95% confidence interval (CI) 1.07-1.85, I²=37%), liver R0 resection rate (RR 2.28, 95% CI 1.34-3.89, I²=0%), ORR (RR 1.20, 95% CI 1.09-1.32, I²=0%), PFS (HR 0.72, 95% CI 0.62-0.84, I²=36%) and OS (HR 0.80, 95% CI 0.66-0.97, I²=0%). There was no significant difference in any Grade $\geq$ 3 adverse event (RR 1.08, 95% CI 0.99-1.17, I²=0%) between two regimens. FOLFOXIRI-Bev was associated with a higher risk of neutropenia (RR 1.77, 95% CI 1.13-2.79, I²=68%) and diarrhea (RR 1.65, 95% CI 1.17-2.32, I²=0%). Triplet chemotherapy plus Bev significantly improved the R0 resection rates, ORR, PFS and OS in comparison with doublet chemotherapy plus Bev in conversion therapy for mCRC patients, with a higher risk of neutropenia and diarrhea.

Partial response after treatment with Conversion chemotherapy: A case report of a patient with colon cancer and unresectable hepatic metastases.

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Many studies have reported radical resection for liver metastasis and the primary tumor could represent an important prognostic factor in patients affected by colorectal liver metastases (CRLM). However, resection of huge liver metastases from colon cancer has been seldom reported. A 58-year-old man presented with huge liver metastases from colon cancer. Laboratory tests revealed elevated tumor markers and a wild-type mutation in the K-RAS gene. A computed tomography scan demonstrated unresectable liver masses with a 16.5-cm maximum diameter and intrahepatic duct dilatation due to compression by the liver metastases. The patient was diagnosed with stage IV descending colon carcinoma with multiple huge hepatic metastases. He was administered 3 treatment courses, including 9 cycles of combined chemotherapy with mFOLFOX6 plus cetuximab (mFOLFOX6 + Cet), and the liver masses reduced. After a preoperative assessment by a multidisciplinary team when the 9 cycles of systemic chemotherapy had been completed, the patient underwent hepatectomy, followed 4 months later by a laparoscopic colectomy. We used a reverse strategy (liver-first) for the patient. In this case, liver-first treatment (systemic chemotherapy of mFOLFOX6 + Cet) was an effective treatment for unresectable CRLM. No postoperative complications occurred. The patient continued to receive postoperative chemotherapy (mFOLFOX6 + Cet) at the latest follow-up. During the 17 months of follow-up, tumor recurrence was un-noted. Treating colorectal cancer patients with huge hepatic metastases is possible, and surgeons should consider various treatment options in the management of these patients.

Successful treatment of conversion chemotherapy for initially unresectable

synchronous colorectal liver metastasis.

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A 72-year-old woman with a sigmoid colon cancer and a synchronous colorectal liver metastasis (CRLM), which involved the right hepatic vein (RHV) and the inferior vena cava (IVC), was referred to our hospital. The metastatic lesion was diagnosed as initially unresectable because of its invasion into the confluence of the RHV and IVC. After she had undergone laparoscopic sigmoidectomy for the original tumor, she consequently had 3 courses of modified 5-fluorouracil, leucovorin, and oxaliplatin (mFOLFOX6) plus cetuximab. Computed tomography revealed a partial response, and the confluence of the RHV and IVC got free from cancer invasion. After 3 additional courses of mFOLFOX6 plus cetuximab, preoperative percutaneous transhepatic portal vein embolization (PTPE) was performed to secure the future remnant liver volume. Finally, a right hemihepatectomy was performed. The postoperative course was uneventful. The patient was discharged from the hospital on postoperative day 13. She had neither local recurrence nor distant metastasis 18 mo after the last surgical intervention. This multidisciplinary strategy, consisting of conversion chemotherapy using FOLFOX plus cetuximab and PTPE, could contribute in facilitating curative hepatic resection for initially unresectable CRLM.

[A successful treatment of conversion chemotherapy by mFOLFOX6 plus cetuximab for initially unresectable synchronous colorectal liver metastases].

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A 63-year-old woman with a synchronous huge colorectal liver metastasis was referred to our institution. The lesion was technically diagnosed unresectable because the estimated future remnant liver volume was insufficient due to the invasion of the three hepatic veins and hepatic hilum. She underwent 7 courses of mFOLFOX6 and 14 administrations of cetuximab as conversion chemotherapy. Periodic abdominal CT scans revealed the tumor becoming PR, and she was free of cancer invasion to the left hepatic vein. After the remainder of chemotherapy lasting 4 weeks, right trisectionectomy and combined partial resection of the inferior vena cava and primary closure was performed. The postoperative course was uneventful and the patient was discharged at 20 days after the operation. She underwent chemotherapy postoperatively, and then underwent laparoscopic sigmoidectomy. A conversion chemotherapy using cetuximab may contribute to ward rapidly reducing tumor size and improving the resectability of initially unresectable huge colorectal liver metastases, thus leading to prolonged survival.

[A Case Report on a Successful Resection after FOLFIRI plus Cetuximab Therapy for Unresectable Colorectal Cancer with Multiple Liver Metastases].

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The patient was a 66-year-old woman with a history of right breast cancer 20 years prior. Her chief complaint was hematochezia, and she was diagnosed as having rectal cancer. She underwent laparoscopic high anterior resection. We made a diagnosis of moderately differentiated adenocarcinoma, type 2, 25×20 mm, pMP, pN0, Stage I, KRAS being wild-type. Multiple liver metastases were detected 6 months after the surgery. Tumor contacted with grison. The tumor was not completely resected as evidenced by the small liver remnant volume. Conversion therapy was administered, and the patient

received 6 courses of FOLFIRI plus cetuximab therapy. Alopecia and grade 1 eruption were observed as adverse effects of the chemotherapy. The tumor size was reduced, and we resected the tumor by performing right lobectomy and partial hepatectomy. At 1 year 3 months after surgery, no recurrence was observed.

Secondary hepatic resection as a therapeutic goal in advanced colorectal cancer.

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Surgery is the only curative option for patients with liver metastases of colorectal cancer, but few patients present with resectable hepatic lesions. Chemotherapy is increasingly used to downstage initially unresectable disease and allow for potentially curative surgery. Standard chemotherapy regimens convert 10%-20% of cases to resectable disease in unselected populations and 30%-40% of those with disease confined to the liver. One strategy to further increase the number of candidates eligible for surgery is the addition of active targeted agents such as cetuximab and bevacizumab to standard chemotherapy. Data from a phase III trial indicate that cetuximab increases the number of patients eligible for secondary hepatic resection, as well as the rate of complete resection when combined with first-line treatment with the FOLFIRI regimen. The safety profiles of preoperative cetuximab or bevacizumab have not been thoroughly assessed, but preliminary evidence indicates that these agents do not increase surgical mortality or exacerbate chemotherapy-related hepatotoxicity, such as steatosis (5-fluorouracil), steatohepatitis (irinotecan), and sinusoidal obstruction (oxaliplatin). Secondary resection is a valid treatment goal for certain patients with initially unresectable liver metastases and an important end point for future clinical trials.
